

How Light Helps Plants Make Food

Audience: teacher

Why light matters

Leaves are often positioned where they can receive light. Light is not the plant's food; it supplies the energy the leaf uses to make food from raw materials.

Do not confuse raw materials with food: Soil supplies water and minerals. The plant makes its food rather than taking ready-made food from the soil.

From light to food

A simple sequence:

- Light reaches the leaf.
- The leaf uses light energy.
- The plant makes food from raw materials.

[Figure pending: A planned diagram showing sunlight reaching a leaf, water entering from the roots, carbon dioxide entering the leaf, and food being made. — Inputs and energy involved when a leaf makes food.]

Evidence from a covered leaf

A controlled comparison of lit and covered leaf areas.

Uncovered | Yes | Present

Covered | No | Absent

Problem: A leaf is partly covered while the rest remains in light. Food-making evidence is later found only in the uncovered part. What conclusion is supported?

- Identify what changed: access to light.
- Check that both parts belong to the same leaf and share other conditions.
- Compare the result: evidence appears in the lit part but not the covered part.
- Connect the changed condition to the different result.

Answer: The comparison supports the conclusion that light is required for the leaf to make food.

Check your understanding

Answer in one or two complete sentences.

q-open-1: Explain why covering part of a leaf helps show that light is needed for the plant to make food.

Which statement best describes the role of soil in plant food making?

- A. Soil is the food the plant eats.
- B. Soil supplies water and minerals, but the plant makes food in its leaves.
- C. Soil supplies sunlight to the roots.

Answer key

q-open-1: The covered part does not receive light while the uncovered part does. When food-making evidence appears only in the lit part, the changed condition shows that light is required.

q-mcq-1: B